DAY – 2 Tasks
Part A — Ownership Investigation (10 Minutes)

Create a directory named ownership_lab.
Inside it, create:
incident_report.txt
network.log
credentials.txt
Display the detailed information for these files.

For each file, identify:
Owner
Group
Owner permissions
Group permissions
Others permissions

: mkdir ownership_lab
cd ownership_lab
touch incident_report.txt network.log credentials.txt
ls -l

```
msis@worker1:~/Desktop/extglob$ mkdir ownership_lab
msis@worker1:~/Desktop/extglob$ cd ownership_lab
msis@worker1:~/Desktop/extglob/ownership_lab$ touch incident_report.txt network.log credentials.txt
msis@worker1:~/Desktop/extglob/ownership_lab$ ls
credentials.txt  incident_report.txt  network.log
msis@worker1:~/Desktop/extglob/ownership_lab$ ls -l
total 0
-rw-rw-r-- 1 msis msis 0 Sep  3 09:44 credentials.txt
-rw-rw-r-- 1 msis msis 0 Sep  3 09:44 incident_report.txt
-rw-rw-r-- 1 msis msis 0 Sep  3 09:44 network.log
msis@worker1:~/Desktop/extglob/ownership_lab$
```

Determine the UID and GID of the current user.
Compare the UID/GID information with the ownership displayed for the newly created files.

: id
id -u
id -g
compare with : ls -ln incident_report.txt network.log credentials.txt

```
msis@worker1:~/Desktop/extglob/ownership_lab$ id
uid=1000(msis) gid=1000(msis) groups=1000(msis),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users),114(lpadmin),125(libvirt)
msis@worker1:~/Desktop/extglob/ownership_lab$ id -u
1000
msis@worker1:~/Desktop/extglob/ownership_lab$ id -g
1000
msis@worker1:~/Desktop/extglob/ownership_lab$ ls -ln incident_report.txt network.log credentials.txt
-rw-rw-r-- 1 1000 1000 0 Sep  3 09:44 credentials.txt
-rw-rw-r-- 1 1000 1000 0 Sep  3 09:44 incident_report.txt
-rw-rw-r-- 1 1000 1000 0 Sep  3 09:44 network.log
msis@worker1:~/Desktop/extglob/ownership_lab$
```

Explain the difference between file ownership and file permissions.
: Ownership determines **which identity/group a permission class applies to**, while permissions determine **what that identity is allowed to do**.

Part B — chown: Changing Ownership (15 Minutes)

Change the owner of incident_report.txt to alice.
Change the owner of network.log to bob.
Change the owner of credentials.txt to charlie.

: sudo chown alice incident_report.txt
sudo chown bob network.log
sudo chown charlie credentials.txt

Verify each change.
: ls -l

```
msis@worker1:~/Desktop/extglob/ownership_lab$ sudo chown alice incident_report.txt
sudo chown bob network.log
sudo chown charlie credentials.txt
[sudo] password for msis:
msis@worker1:~/Desktop/extglob/ownership_lab$ ls -l
total 0
-rw-rw-r-- 1 charlie msis 0 Sep  3 09:44 credentials.txt
-rw-rw-r-- 1 alice   msis 0 Sep  3 09:44 incident_report.txt
-rw-rw-r-- 1 bob     msis 0 Sep  3 09:44 network.log
msis@worker1:~/Desktop/extglob/ownership_lab$
```

Try changing the owner of a file to another user without using sudo. Record the result.
: chown alice network.log

```
msis@worker1:~/Desktop/extglob/ownership_lab$ chown alice network.log
chown: changing ownership of 'network.log': Operation not permitted
msis@worker1:~/Desktop/extglob/ownership_lab$
```

Explain why an ordinary user is generally not permitted to transfer file ownership to another user.

: An ordinary user generally cannot give their file away to another user because unrestricted ownership transfer could be abused to bypass accountability, security controls, or other administrative policies.

Using a single command, configure:
incident_report.txt
Owner : alice
Group : security
Verify the resulting ownership.

: sudo chown alice:security incident_report.txt
ls -l

```
msis@worker1:~/Desktop/extglob/ownership_lab$ sudo chown alice:security incident_report.txt
msis@worker1:~/Desktop/extglob/ownership_lab$ ls -l
total 0
-rw-rw-r-- 1 charlie msis     0 Sep  3 09:44 credentials.txt
-rw-rw-r-- 1 alice   security 0 Sep  3 09:44 incident_report.txt
-rw-rw-r-- 1 bob     msis     0 Sep  3 09:44 network.log
msis@worker1:~/Desktop/extglob/ownership_lab$
```

Part C — chgrp: Changing Group Ownership (15 Minutes)

Change the group ownership of network.log to developers.
Change the group ownership of credentials.txt to security.
Verify the changes.

: sudo chgrp developers network.log
sudo chgrp security credentials.txt
ls -l

```
msis@worker1:~/Desktop/extglob/ownership_lab$ sudo chgrp developers network.log
sudo chgrp security credentials.txt
msis@worker1:~/Desktop/extglob/ownership_lab$ ls -l
total 0
-rw-rw-r-- 1 charlie security   0 Sep  3 09:44 credentials.txt
-rw-rw-r-- 1 alice   security   0 Sep  3 09:44 incident_report.txt
-rw-rw-r-- 1 bob     developers 0 Sep  3 09:44 network.log
msis@worker1:~/Desktop/extglob/ownership_lab$
```

Display the group memberships of alice, bob, and charlie.
: groups alice
groups bob
groups charlie

```
msis@worker1:~/Desktop/extglob/ownership_lab$ groups alice
groups bob
groups charlie
alice : alice security
bob : bob security developers
charlie : charlie security
msis@worker1:~/Desktop/extglob/ownership_lab$
```

Log in as alice and create a file named alice_report.txt.
Attempt to change the group of alice_report.txt to security.
Attempt to change its group to developers.
Compare the results.
Explain how a user's group membership affects their ability to change the group ownership of a file they own.
: su – alice
touch alice_report.txt
ls -l
chgrp security alice_report.txt
chgrp developers alice_report.txt

```
msis@worker1:~/Desktop/extglob/ownership_lab$ su - alice
Password:
$ cd /path/to/ownership_lab
-sh: 1: cd: can't cd to /path/to/ownership_lab
$ ls
$ touch alice_report.txt
$ ls
alice_report.txt
$ ls -l
total 0
-rw-rw-r-- 1 alice alice 0 Sep  3 11:05 alice_report.txt
$ chgrp security alice_report.txt
$ chgrp developers alice_report.txt
chgrp: changing group of 'alice_report.txt': Operation not permitted
$
```

Part D — Ownership + Permission Investigation (20 Minutes)

Configure:

-rw-r----- alice security incident_report.txt
Identify what operations are available to:
alice
charlie
bob

: sudo chown alice:security incident_report.txt
sudo chmod 640 incident_report.txt
ls -l

A user who belongs to neither security nor the file owner
charlie belongs to security. Test whether Charlie can:
Read incident_report.txt
Modify incident_report.txt
Delete incident_report.txt
: File deletion is controlled primarily by the **parent directory's permissions**, not by the permissions displayed on the file itself.
If the directory is owned by your original user and is not writable by Charlie/Bob, they won't be able to delete the file regardless of the file's own rw permissions.

```
-rw-rw-r-- 1 charlie security   0 Sep  3 09:44 credentials.txt
-rw-r----- 1 alice   security   0 Sep  3 09:44 incident_report.txt
-rw-rw-r-- 1 bob     developers 0 Sep  3 09:44 network.log
msis@worker1:~/Desktop/extglob/ownership_lab$ su - charlie
Password:
$ cat incident_report.txt
cat: incident_report.txt: No such file or directory
$ exit
msis@worker1:~/Desktop/extglob/ownership_lab$ sudo chown alice incident_report.txt
sudo chown bob network.log
sudo chown charlie credentials.txt
msis@worker1:~/Desktop/extglob/ownership_lab$ sudo chgrp developers network.log
sudo chgrp security credentials.txt
msis@worker1:~/Desktop/extglob/ownership_lab$ sudo chown alice:security incident_report.txt
sudo chmod 640 incident_report.txt
msis@worker1:~/Desktop/extglob/ownership_lab$ su - charlie
Password:
$ ls
$ touch incident_report.txt
$ cat incident_report.txt
$ echo "Charlie was here" >> incident_report.txt
$ rm incident_report.txt
$ exit
msis@worker1:~/Desktop/extglob/ownership_lab$ sudo chmod 660 incident_report.txt
msis@worker1:~/Desktop/extglob/ownership_lab$ ls -l incident_report.txt
-rw-rw---- 1 alice security 0 Sep  3 09:44 incident_report.txt
msis@worker1:~/Desktop/extglob/ownership_lab$ cat incident_report.txt
cat: incident_report.txt: Permission denied
msis@worker1:~/Desktop/extglob/ownership_lab$ su - charlie
Password:
$ cat incident_report.txt
cat: incident_report.txt: No such file or directory
$ touch incident_report.txt
$ cat incident_report.txt
$ echo "Charlie modified the file" >> incident_report.txt
$ cat incident_report.txt
Charlie modified the file
$ exit
msis@worker1:~/Desktop/extglob/ownership_lab$ groups bob
bob : bob security developers
msis@worker1:~/Desktop/extglob/ownership_lab$ sudo gpasswd -d bob security
Removing user bob from group security
msis@worker1:~/Desktop/extglob/ownership_lab$ groups bob
bob : bob developers
msis@worker1:~/Desktop/extglob/ownership_lab$
```